

Numerical Analysis chapter 3 theoretical homework

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I

Proof:

$$p(1) = 1, p'(1) = -3, p''(1) = 6, s(0) = 0$$

We assume that $p(x) = a_0 + a_1x + a_2x^2 + a_3x^3$. We have: $a_0 = 0, a_1 = 12, a_2 = -18, a_3 = 7$ $p''(0) = -36 \neq 0$ So $s(x)$ is not a natural cubic spline.

II

a

If we want to determine s uniquely, we need to have $3 * (n - 1) = 3n - 3$ conditions. From $s_i = f_i$, we have n conditions. From $s'_i = s'_{i+1}$, we have $n - 2$ conditions. From $s''_i = s''_{i+1}$, we have $n - 2$ conditions. We have $3n - 4$ conditions in total, so s can not be uniquely determined.

b

$$m_i = s'(x_i), p_i = s|_{[x_i, x_{i+1}]}, p_i(x_i) = f(x_i), p_i(x_{i+1}) = f(x_{i+1})$$

$$\begin{aligned} p_i(x) &= f_i + m_i(x - x_i) + \frac{f[x_i, x_{i+1}] - m_i}{x_{i+1} - x_i}(x - x_i)^2 \\ &= f_i + m_i(x - x_i) + \frac{\frac{f_{i+1} - f_i}{x_{i+1} - x_i} - m_i}{x_{i+1} - x_i}(x - x_i)^2 \\ &= f_i + m_i(x - x_i) + \frac{f_{i+1} - f_i - m_i(x_{i+1} - x_i)}{(x_{i+1} - x_i)^2}(x - x_i)^2 \end{aligned}$$

c

$$\begin{aligned} p'_i &= m_i + 2(x - x_i) \frac{f_{i+1} - f_i - m_i(x_{i+1} - x_i)}{(x_{i+1} - x_i)^2} \\ p'_{i+1} &= m_{i+1} + 2(x - x_{i+1}) \frac{f_{i+2} - f_{i+1} - m_{i+1}(x_{i+2} - x_{i+1})}{(x_{i+2} - x_{i+1})^2} \end{aligned}$$

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$$p_i'(x_{i+1}) = p_{i+1}'(x_{i+1})$$

$$m_{i+1} = m_i + 2 \frac{f_{i+1} - f_i - m_i(x_{i+1} - x_i)}{(x_{i+1} - x_i)}$$

III

$$\begin{cases} s_1(x) = 1 + c(x+1)^3, \\ s_1'(x) = 3c(x+1)^2, \\ s_2''(x) = 6c(x+1), \end{cases} \quad \begin{cases} s_2(0) = 1 + c, \\ s_2'(0) = 3c, \\ s_2''(0) = 6c, \\ s_2(1) = 0. \end{cases}$$

We have $s_2(x) = 1 + c + 3cx + 3cx^2 - cx^3$ and $s_2(1) = -1$, so $c = -\frac{1}{3}$.

IV

$$f(x) = \cos\left(\frac{\pi}{2}x\right), x \in [-1, 1]$$

a

$$\begin{cases} f(x) = \cos\left(\frac{\pi}{2}x\right) \\ f'(x) = -\frac{\pi}{2}\sin\left(\frac{\pi}{2}x\right) \\ f''(x) = -\frac{\pi^2}{4}\cos\left(\frac{\pi}{2}x\right) \end{cases}$$

We have $M_1 = s''(-1) = 0, s''(1) = 0 = M_3$ and we know $f(-1) = 0, f(1) = 0, f(0) = 1$ From **lemma 3.4**, we have

$$\mu_2 M_1 + 2M_2 + \lambda_2 M_3 = 6f[-1, 0, 1].$$

It is trivial that $f[-1, 0, 1] = -1, \mu_2 = \frac{1}{2}, \lambda_2 = \frac{1}{2}$. So, we have $M_2 = -3$.

$$\begin{cases} s''(0) = -3 \\ s''(1) = 0 \\ s''(-1) = 0. \end{cases}$$

We have

$$s(x) = \begin{cases} -\frac{1}{2}x^3 - \frac{3}{2}x^2 + 1, & \text{if } x \in [-1, 0] \\ \frac{1}{2}x^3 - \frac{3}{2}x^2 + 1, & \text{if } x \in [0, 1] \end{cases}$$

b

$$g(-1) = 0, g(0) = 1, g(1) = 0$$

i

It is obvious that $g(x) = -x^2 + 1$.

$$\begin{aligned}\int_{-1}^1 [s''(x)]^2 dx &= \int_{-1}^0 (-3x - 3)^2 dx + \int_0^1 (3x - 3)^2 dx = 6 \\ \int_{-1}^1 [g''(x)]^2 dx &= 8 > \int_{-1}^1 [s''(x)]^2 dx\end{aligned}$$

ii

$$\int_{-1}^1 [f''(x)]^2 dx = \frac{\pi^4}{16} > 6$$

V

a

$$\begin{aligned}B_i^0 &= \begin{cases} 1, & \text{if } x \in [t_{i-1}, t_i] \\ 0, & \text{otherwise} \end{cases} \\ \hat{B}_i(x) &= \begin{cases} \frac{x-t_{i-1}}{t_i-t_{i-1}}, & x \in [t_{i-1}, t_i] \\ \frac{t_{i+1}-x}{t_{i+1}-t_i}, & x \in [t_i, t_{i+1}] \\ 0, & \text{otherwise} \end{cases} \\ B_i^{n+1} &= \frac{x-t_{i-1}}{t_{i+n}-t_{i-1}} B_i^n(x) + \frac{t_{i+n+1}-x}{t_{i+n+1}-t_i} B_{i+1}^n(x) \\ B_i^2(x) &= \begin{cases} \frac{(x-t_{i-1})^2}{(t_{i+1}-t_{i-1})(t_i-t_{i-1})}, & x \in (t_{i-1}, t_i] \\ \frac{(x-t_{i-1})(t_{i+1}-x)}{(t_{i+1}-t_{i-1})(t_{i+1}-t_i)} + \frac{(t_{i+2}-x)(x-t_i)}{(t_{i+2}-t_i)(t_{i+1}-t_i)}, & x \in (t_i, t_{i+1}] \\ \frac{(t_{i+2}-x)^2}{(t_{i+2}-t_i)(t_{i+2}-t_{i+1})}, & x \in (t_{i+1}, t_{i+2}] \\ 0, & \text{otherwise.} \end{cases}\end{aligned}$$

b

$$\frac{d}{dx} B_i^2(x) = \begin{cases} \frac{2(x-t_{i-1})}{(t_{i+1}-t_{i-1})(t_i-t_{i-1})}, & x \in (t_{i-1}, t_i] \\ \frac{t_{i+1}-2x+t_{i+1}}{(t_{i+1}-t_{i-1})(t_{i+1}-t_i)} + \frac{t_{i+2}-2x+t_i}{(t_{i+2}-t_i)(t_{i+1}-t_i)}, & x \in (t_i, t_{i+1}] \\ \frac{-2(t_{i+2}-x)}{(t_{i+2}-t_i)(t_{i+2}-t_{i+1})}, & x \in (t_{i+1}, t_{i+2}] \\ 0, & \text{otherwise.} \end{cases}$$

We have

$$\begin{aligned}\lim_{x \rightarrow t_i^+} \left(\frac{t_{i+1}-2x+t_{i+1}}{(t_{i+1}-t_{i-1})(t_{i+1}-t_i)} + \frac{t_{i+2}-2x+t_i}{(t_{i+2}-t_i)(t_{i+1}-t_i)} \right) &= \frac{2}{t_{i+1}-t_{i-1}} \\ \lim_{x \rightarrow t_{i+1}^+} \left(\frac{-2(t_{i+2}-x)}{(t_{i+2}-t_i)(t_{i+2}-t_{i+1})} \right) &= \frac{-2}{t_{i+2}-t_i}\end{aligned}$$

So $B_i^2(x)$ is C^1 continuous.

c

From the deritive of $B_i^2(x)$, we have $B_i^2(x)$ increase from t_{i-1} to t_i and decrease from t_i to t_{i+1} . We have $B_i^2(t_i) > 0$ and $B_i^2(t_{i+1}) < 0$ in $[t_{i-1}, t_{i+1}]$. From the continuity of $\frac{d}{dx} B_i^2(x)$, we have $B_i^2(x)$ has a zero point in $[t_{i-1}, t_{i+1}]$.

d

We assume that the zero point of $\frac{d}{dx} B_i^2(x)$ is x^* .

$$\frac{d}{dx} B_i^2(x) = \begin{cases} > 0, & x \in (t_{i-1}, x^*) \\ = 0, & x = x^* \\ < 0, & x \in (x^*, t_{i+2}) \\ 0, & \text{otherwise} \end{cases}$$

So $\max B_i^2(x^*) = \frac{t_{i+2}-t_{i-1}}{t_{i+2}-t_{i-1}+t_{i+1}-t_i} < 1$ and $\min B_i^2(x^*) = 0$. So $B_i^2(x) \in [0, 1]$

e

Let $t_i = i$ we have

$$B_i^2(x) = \begin{cases} \frac{(x-i+1)^2}{2}, & x \in (i-1, i] \\ \frac{(x-i+1)(-x+i+1)+(x-i)(-x+i+2)}{2}, & x \in (i, i+1] \\ \frac{(-x+i+2)^2}{2}, & x \in (i+1, i+2] \\ 0, & \text{otherwise.} \end{cases}$$

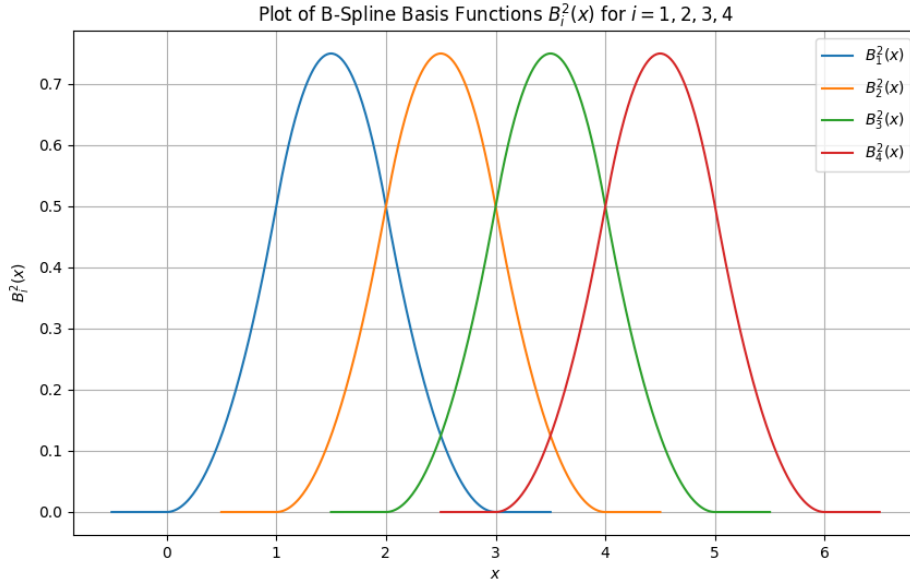


Figure 1: The graph of $B_i^2(x)$

VI

$$B_i^n(x) = (t_{i+n} - t_{i-1})[t_{i-1}, \dots, t_{i+n}](t-x)_+^n, \forall n$$

When $n = 2$, $B_i^2(x) = (t_{i+2} - t_{i-1})[t_{i-1}, \dots, t_{i+2}](t-x)_+^2$.

$$B_i^{n+1} = \frac{x - t_{i-1}}{t_{i+n} - t_{i-1}} B_i^n(x) + \frac{t_{i+n+1} - x}{t_{i+n+1} - t_i} B_{i+1}^n(x)$$

$$B_i^0(x) = (t_i - t_{i-1})[t_{i-1}, t_i](t-x)_+^0$$

$$\therefore B_i^1(x) = (x - t_{i-1})[t_{i-1}, t_i](t-x)_+^0 + (t_{i+1} - x)[t_i, t_{i+1}](t-x)_+^0$$

$$(t_{i-1} \cdots t_{i+n})_+^{n+1} = (t_{i-1} - x)[t_{i-1}, \dots, t_{i+n}](t-x)_+^n$$

$$\implies B_i^1(x) = (t_{i+1} - t_{i-1})[t_{i-1}, t_i, t_{i+1}](t-x)_+^n$$

Similarly, we have

$$\begin{aligned} B_i^2(x) &= (x - t_{i-1})[t_{i-1}, t_i, t_{i+1}](t-x)_+^1 + (t_{i+2} - x)[t_i, t_{i+1}, t_{i+2}](t-x)_+^1 \\ &= -[t_{i-1}, t_i, t_{i+1}](t-x)_+^2 + [t_i, t_{i+1}, t_{i+2}](t-x)_+^2 \\ &= (t_{i+2} - t_{i-1})[t_{i-1}, t_i, t_{i+1}, t_{i+2}](t-x)_+^2 \end{aligned}$$

The proof is completed.

VII

$$\begin{aligned} \frac{d}{dx} B_i^n(x) &= n \left(\frac{B_i^{n-1}(x)}{t_{i+n-1} - t_{i-1}} - \frac{B_{i+1}^{n-1}(x)}{t_{i+n} - t_i} \right) \\ \int_{t_{i-1}}^{t_{i+n}} \frac{d}{dx} B_i^n(x) dx &= 0 \end{aligned}$$

This is because the derivative of $B_i^n(x)$ is continuous and $B_i^n(t_{i-1}) = B_i^n(t_{i+n}) = 0$

$$\therefore \int_{t_{i+n+1}}^{t_{i-1}} \frac{d}{dx} B_i^n(x) dx \implies \int_{t_{i-1}}^{t_{i+n}} \frac{B_i^{n-1}(x)}{t_{i+n-1} - t_{i-1}} dx = \int_{t_i}^{t_{i+n}} \frac{B_{i+1}^{n-1}(x)}{t_{i+n} - t_i} dx \square$$

VIII

$$\tau_{m-n}(x_i \cdots x_{i+n}) = [x_i, x_{i+1}, \dots, x_{i+n}] x^m$$

a

$$\tau_2(x_i \cdots x_{i+2}) = [x_i, x_{i+1}, x_{i+2}] x^4$$

$$\begin{array}{c|c} x_i & x_i^4 \\ x_{i+1} & x_{i+1}^4 \quad (x_{i+1}^3 + x_i x_{i+1}^2 + x_i^2 x_{i+1} + x_i^3) \\ x_{i+2} & x_{i+2}^4 \quad (x_{i+2}^3 + x_{i+1} x_{i+2}^2 + x_{i+1}^2 x_{i+2} + x_{i+1}^3) \quad x_i^2 + x_{i+1}^2 + x_{i+2}^2 + x_{i+1} x_{i+2} + x_i x_{i+1} + x_i x_{i+2} \end{array}$$

The proof is completed.

b

$$\begin{aligned}(x_{n+1} - x_1)\tau_k(x_1 \cdots x_{n+1}) &= \tau_{k+1}(x_1 \cdots x_{n+1}) - \tau_{k+1}(x_1 \cdots x_n) - x_1\tau_k(x_1 \cdots x_{n+1}) \\ &= \tau_{k+1}(x_2 \cdots x_{n+1}) - \tau_{k+1}(x_1 \cdots x_n)\end{aligned}$$

$$n = 0 : \tau_m(x_i) = [x_i]x^m$$

$$k = n + 1 :$$

$$\begin{aligned}\tau_{m-n-1}(x_i \cdots x_{i+n+1}) &= \frac{\tau_{m-n}(x_{i+1} \cdots x_{i+n+1}) - \tau_{m-n}(x_i \cdots x_{i+n})}{x_{i+n+1} - x_i} \\ &= \frac{[x_{i+1}, x_{i+2} \cdots x_{i+n+1}]x^m - [x_i \cdots x_{i+n}]x^m}{x_{i+n+1} - x_i} \\ &= [x_i, x_{i+1} \cdots x_{i+n+1}]x^m\end{aligned}$$

The proof is completed.[1]

Acknowledgement

References

- [1] OpenAI. *ChatGPT: A Conversational Agent based on GPT-4 Architecture*. <https://chat.openai.com>. Accessed: 2024-11-04.